Homework 1

Events, counting, and conditional probability

60 points. Six problems, 10 points each. Due October 6 at 10 p.m. Pacific; final no-penalty grace cutoff October 7 at 10 p.m. Pacific. Handwrite your work and submit one legible PDF in Gradescope, mapping pages to all six questions. Paper or stylus handwriting is accepted, subject to approved accommodations. Show reasoning, support and assumptions. Write independently and disclose collaborators. Do not use AI tools to generate, solve, check, or rewrite assessed work. Exact expressions are accepted unless stated otherwise.

1. Events in a service log — 10 points

For a randomly selected request, let A mean slow and B mean retried. Assume $P(A) = 0.30$, $P(B) = 0.25$, and $P(A \cap B) = 0.10$.

(a) [5] Express "slow or retried" and "neither" as events, and find their probabilities.

(b) [5] Express "exactly one of slow and retried" as a disjoint union, and find its probability. Explain why $P(A) + P(B)$ is not the answer.

2. Roles and subsets — 10 points

Eight distinct students volunteer. No student can hold more than one role.

(a) [5] How many ways can a coordinator, recorder, and presenter be assigned? Explain each factor.

(b) [5] How many three-person teams can be chosen? If all such teams are equally likely, what is the probability that a specified student is included?

3. Allocation with and without collisions — 10 points

Three labeled jobs independently choose one of five labeled servers, each uniformly. A collision means at least two jobs choose the same server.

(a) [5] Specify the sample space and the probability of one outcome. Find the probability of no collision.

(b) [5] Find the probabilities of a collision and of all three jobs choosing the same server. Are these events identical?

4. Two observations of a die experiment — 10 points

Two independent fair six-sided dice are rolled. Let S be their sum.

(a) [5] Find $P(S = 9 \mid \text{at least one die is 6})$. List the favorable ordered pairs and count the conditioning outcomes.

(b) [5] Find $P(S = 9 \mid \text{first die is 6})$ and $P(S = 9 \mid \text{exactly one die is 6})$. Explain the different denominators.

5. Mixture of production sources — 10 points

A device comes from source B_1, B_2, or B_3 with probabilities 0.50, 0.30, and 0.20. Its conditional failure probabilities are 0.02, 0.04, and 0.10, respectively. Let D denote failure.

(a) [5] Compute $P(D \cap B_i)$ for each source and then $P(D)$. State the partition used.

(b) [5] Given that the device failed, find $P(B_3 \mid D)$. Compare this with the prior source share.

6. An alert and the base rate — 10 points

In an idealized screening system, $P(G) = 0.02$, $P(L \mid G) = 0.90$, and $P(L \mid G^c) = 0.05$, where G is a genuine fault and L is an alert.

(a) [5] Find the alert probability and the probability of a genuine fault given an alert.

(b) [5] Keep the two conditional alert probabilities fixed but change $P(G)$ to 0.20. Recompute the posterior and explain the change using model proportions in 1,000 cases, without assuming observed counts equal those proportions.